

3.3 Rational Exponents – Worksheet

MCR3U

1) Evaluate each cube root.

a) $\sqrt[3]{64}$

b) $(-1000)^{\frac{1}{3}}$

c) $\sqrt[3]{\frac{1}{8}}$

d) $\left(\frac{8}{27}\right)^{\frac{1}{3}}$

2) Evaluate each root.

a) $81^{\frac{1}{4}}$

b) $\sqrt[4]{\frac{16}{625}}$

c) $64^{\frac{1}{6}}$

d) $\sqrt[5]{-100\,000}$

3) Evaluate.

a) $8^{\frac{2}{3}}$

b) $32^{\frac{4}{5}}$

c) $(-64)^{\frac{5}{3}}$

d) $\left(\frac{1}{10\,000}\right)^{\frac{3}{4}}$

4) Evaluate.

a) $16^{-\frac{1}{4}}$

b) $25^{-\frac{3}{2}}$

c) $\left(\frac{1}{8}\right)^{-\frac{7}{3}}$

d) $\left(-\frac{1}{32}\right)^{-\frac{2}{5}}$

e) $\left(\frac{10\,000}{81}\right)^{-\frac{3}{4}}$

f) $\left(-\frac{8}{27}\right)^{-\frac{2}{3}}$

5) Simplify. Express your answers using only positive exponents.

a) $x^{\frac{1}{4}} \cdot x^{\frac{1}{4}}$

b) $\left(m^{\frac{1}{3}}\right)\left(m^{\frac{3}{4}}\right)$

c) $\frac{w^{\frac{1}{2}}}{\frac{1}{w^{\frac{2}{3}}}}$

d) $\frac{ab^2}{\frac{1}{a^2b^3}}$

e) $\left(y^{\frac{1}{2}}\right)^{\frac{2}{3}}$

f) $\left(u^{\frac{3}{4}}v^{\frac{1}{2}}\right)^{\frac{2}{9}}$

6) Simplify. Express your answers using only positive exponents.

a) $k^{\frac{3}{4}} \div k^{-\frac{1}{4}}$

b) $\frac{p^{-\frac{2}{3}}}{p^{\frac{5}{6}}}$

c) $\left(y^{\frac{2}{3}}\right)^{-3}$

d) $\left(w^{-\frac{8}{9}}\right)^{-\frac{3}{4}}$

e) $(8x)^{\frac{2}{3}}(27x)^{-\frac{1}{3}}$

f) $5\left(7y^{-\frac{2}{3}}\right)^{-2}$

7) The surface area, S , of a sphere can be expressed in terms of its volume, V , using the formula

$S(V) = (4\pi)^{\frac{1}{3}}(3V)^{\frac{2}{3}}$. A beach ball has a volume of 24 000 cm³. Find its surface area, to the nearest hundred square centimeters.